

Paper 5 (fisher_path_speed v2) — corrections

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1. Abstract — topology / CV wording

Issue: CV 0.14–0.17 in the abstract did not match Snake CV ≈ 0.23 in Table 1 ($K=7$). Also clarify that topology stability is stated per fixed K .

Within each SAX alphabet size K in $\{5,7,9\}$, the product $A\tau$ is stable across the three Sycamore hardware topologies among readouts where the exponential model wins: at $K=7$ the topology-wise medians of $A\tau$ agree within $\sim 2\%$ and the topology-level coefficients of variation (CV) range from 0.13 (Bulk) to 0.23 (Snake; Table 1).

The same qualitative pattern holds at $K=5$ and $K=9$ (exponential-fit subset): per-topology medians differ by only a few percent, with CV values ~ 0.07 – 0.22 when CV is computed on $A\tau$ within each topology.

Varying K across $\{5,7,9\}$, the cross- K medians of $A\tau$ follow

$A\tau \sim 21.5 * K^{0.57}$ with residuals below 1.4% at the three tabulated K values.

(LaTeX version with math mode is in the repository script output if you prefer: use $K \in \{5,7,9\}$, $A \times \tau$, etc.)

2. Table 2 — replace HTML n_{exp}

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\begin{tabular}{@{}lcccccc@{}}
\toprule
 $K$  &  $d=K(K-1)$  &  $n_{\text{exp}}$  & Median  $A\tau$  & Pred.  $\tau$  & Error \% \\
\midrule
5 & 20 & 26 & 53.32 & 53.64 & 0.6\% \\
7 & 42 & 25 & 65.87 & 64.94 & 1.4\% \\
9 & 72 & 25 & 74.30 & 74.90 & 0.8\% \\
\bottomrule
\end{tabular}
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3. §2.1 / §2.2 — sentence for $d = K(K-1)$

Here $d = K(K-1)$ counts the total number of free coordinates in the row-stochastic model: each of the K rows lies on a $(K-1)$ -simplex, and the Fisher-Rao metric is taken with respect to the product of these K simplices (block-diagonal Fisher information).

4. §3.3 — after Table 1 (numeric support for $K=5,9$)

For $K=5$ (26 exponential-fit readouts) the topology-level medians of $A\tau$ are 54.13 (Snake, $n=7$), 49.00 (Block, $n=13$), and 56.79 (Bulk, $n=6$), with CV = std/median within each topology of 0.22, 0.16, and 0.11 respectively. For $K=9$ (25 exponential-fit readouts) the corresponding medians are 75.31, 73.50, and 74.30 ($n=7,13,5$) with CV 0.16, 0.17, and 0.07. Thus the "topology independence within fixed K " statement is not special to $K=7$; it is reproducible across alphabet sizes, with Snake rows typically slightly noisier than Bulk in this CV sense.

5. §3.5 LLM — replace “10 non-pathological”

The same analysis applied to LLM entropy time series (Qwen2.5-Coder:7B, $K=7$, twelve experimental conditions) yields noisier profiles: the power-law tail model is preferred by AIC in all twelve conditions, consistent with multi-scale temporal structure documented in the Curvature Research Note.

We therefore do not interpret the exponential sub-parameters (τ , A , hence $A\tau$) as primary estimands for LLMs; the reported median $A\tau \sim 91.5$ (and $p = 0.11$ vs. 65.97) is exploratory only and is not used in the main Sycamore scaling analysis.

6. Summary bullet 3 (replace)

\item The product $A\tau$ is stable across hardware topologies within each fixed K among exponential-fit readouts (Table 1 at $K=7$; qualitatively similar at K in $\{5,9\}$ as tabulated in the repository). Across K in $\{5,7,9\}$, median $A\tau$ scales as $21.5 * K^{0.57}$ (residuals $< 1.4\%$).

LaTeX math check (compiles)

Table header: n_{exp} (not n_{exp}).

Degrees of freedom: $d = K(K - 1)$.

Power law: $A \times \tau \approx 21.5 K^{0.57}$.